

Program : Diploma in Polymer Technology	
Course Code : 5093A	Course Title: Injection Mould Design
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To acquaint with various metals and operations carried out in plastic injection mould making.
- To assimilate design and working principles of various components of plastic injection moulds.

Course Prerequisites:

Topic	Course code	Course name	Semester
Orthographic projections - Pictorial		Engineering Graphics	1
Machine shop and basic fitting operations		Engineering Workshop Practice	1 & 2
Metals and alloys		Basic Mechanical Engineering	3
Plastic materials for Injection moulding, Injection machine and clamping unit		Plastic Technology	4

Course Outcome:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate appropriate metals and machining operations for injection mould making	15	Applying
CO2	Explain various parts of an injection mould and operations involved in their construction	15	Understanding
CO3	Describe injection mould feed system and ejection system	14	Applying
CO4	Generalize parting surface and cooling systems of	14	Applying

	injection mould		
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3				
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate appropriate metals and machining operations for injection mould making		
M1.01	Summarize metals and alloys with main focus on steel	2	Remembering
M1.02	Familiarize injection mould making steel	2	Understanding
M1.03	Illustrate important machine tools and their operations used in Injection mould making	9	Understanding
M1.04	Outline machining operations required for a mould plate preparation	2	Applying
Contents: Classification of metals – ferrous and non-ferrous. Composition of plain carbon steel. Alloy steel – Mould steel-types – requirements. List important machine tools used for injection mould making. Identify parts of a Lathe using line diagram. Lathe operations in brief; turning (different types), facing, parting off, forming, thread cutting, knurling, drilling, reaming, boring. Mode of operation of planer and shaper. Comparison between planer and shaper. Cylindrical grinding machine – sketch and mode of operation. Basic machine tool operations for manufacturing a simple mould plate.			
CO2	Explain various parts of an injection mould and operations involved in their construction		
M2.01	Know the injection mould clamping unit and injection mould general terms	2	Remembering

M2.02	Illustrate core and cavity making techniques	6	Understanding
M2.03	Familiarize different parts of an ideal injection mould, ancillary parts	6	Understanding
M2.04	Outline steps involved in Bench fitting of Injection mould.	1	Understanding
	Series Test – I	1	

Contents:

Understand the terms; Injection mould clamping unit - fixed & moving halves, tie bar & bush. Terms relating Injection moulds - impression, cavity and core.

Cavity and core making techniques; Casting, Electro - deposition, Cold hobbing, Pressure casting. Integer and insert type mould. Bolsters – requirements and classification. Types of cavity and core inserts.

Sketch of an ideal injection mould and label its parts. Ancillary items; Guide bush & Guide pillar – functions, types, location and positioning. Sprue bush – function and types. Register ring – function and types.

Steps involved in Bench fitting of an Injection mould.

CO3	Describe injection mould feed system and ejection system		
M3.01	Identify Injection mould feed system	2	Remembering
M3.02	Illustrate runner and gate in terms of types and designs	6	Understanding
M3.03	Explain different parts of an Ejector grid and Ejector plate assembly	3	Understanding
M3.04	Illustrate the ejector plate actuation-return mechanisms and ejection techniques	3	Understanding

Contents:

Define feed system – sprue, runner and gate. Runner cross-sectional shapes and efficiency. Runner size and layouts. Gates – Need for small cross-sectional area, Position of gate, Balanced gating, Types of gates.

Ejection – Ejector grid and designs, Ejector plate assembly - Ejector plate & Retaining plate, Ejector plate actuation methods, Guiding and supporting of ejector plate assembly, Ejector rod and ejector rod bush – designs. Ejector plate assembly return systems, Stop pins. Ejection techniques – different types, Ejection from fixed half. Sprue pullers – Type A & B.

CO4	Generalize parting surface and cooling systems of injection mould		
M4.01	Illustrate various parting surface and relief	2	Understanding
M4.02	Explain the significance of mould surface balancing and venting	2	Understanding

M4.03	Explain various cooling systems adopted in Injection moulds	8	Understanding
M4.04	Explain the water connections and seals used in mould cooling	2	Understanding
	Series Test – II	1	
Contents: Parting surface – Classification, Non-flat parting surfaces – Stepped, Profiled, Angled & Complex edge forms. Balancing of mould surfaces, Relief of parting surfaces, Mould venting. Mould cooling - Factors deciding mould temperature. Types of cooling systems – Integer mould plates, Integer core plates, Insert-Bolster assembly, Cooling other mould parts – Stripper plate, Valve type ejector head, Sprue bush. Water connections and Seals in brief –O-ring seals.			

Text / Reference:

T/R	Book Title/Author
T1	Elements of Workshop Technology – Vol. I & II – S K Hajrah Choudhury (Media Promoters and Publishers Pvt. Ltd.)
T2	Injection Mould Design- R.G.W.Pye (Publisher – John Wiley & Sons Inc.)
T3	Injection Molding Handbook - Dominick V. Rosato(Publisher – Springer)
R1	How to make injection molds - Georg Menges (Publisher - Oxford University Press, USA)
R2	Handbook of thermoplastics injection mould design – P S Cracknell(Publisher - Blackie Academic & Professional)
R3	Injection Mold Design Engineering - David Kazmer (Publisher – Hanser)

Online Resources:

Sl.No	Website Link
1	https://www.dme.net
2	https://techcenter.lanxess.com
3	https://www.rtpcompany.com